

Loose-Screw Activation Framework: Learnable Leak as a Controlled Auxiliary Signal Path

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Transparency and Provenance Note

This document is an early research draft built around a simple personal idea: leak in an activation function need not be treated only as an error to suppress, but may be useful as a bounded and learnable auxiliary signal path. The author does not claim prior expertise across the full activation-function literature and presents this work as an exploratory empirical study rather than as a settled scientific result.

Parts of the brainstorming, code scaffolding, analysis, and manuscript drafting were produced with the assistance of large language models, including ChatGPT, Gemini, and Grok, and were then selected, checked, and edited by the author. The purpose of including this note is transparency: the underlying research direction and framing came from the author, while LLMs were used as exploratory and writing tools to help turn that idea into runnable experiments and a readable draft.

For additional transparency, the author provides the originating LLM conversations and notebook used for the experiments. The following are the final links:

- Originating ChatGPT conversation:
<https://chatgpt.com/share/69ecc758-6b70-8324-9130-1fb74da021d8>
- Originating Gemini conversation:
<https://gemini.google.com/share/ed4e13fe4180>
- Originating Grok conversation:
https://grok.com/share/bGVnYWN5LWNvcHk_f5f1283e-72df-4871-9b7c-c4d6abcccce15
- Experiment notebook / Colab:
https://colab.research.google.com/drive/1HFgLXHdaQ6V50xKWQqT2H_tq0WERzhMR?usp=sharing

This manuscript should not be interpreted as a peer-reviewed claim of novelty, priority, or superior performance. Readers should independently verify the experiments, compare against prior work, and treat the present results as an initial proof-of-concept study.

Abstract

We study a family of neural activations built around a simple idea: leak need not be treated only as an error to suppress, but can be used as a bounded auxiliary signal path. The main formulation is a mixed dual-path activation in which a clean ReLU branch is combined with a bounded pressure-aware leak branch through learnable path weights. Across the current real-dataset benchmarks, the mixed formulation is competitive in several settings and consistently maintains very low dead-unit rates, but it does not uniformly outperform standard baselines. A focused leak-band sweep further suggests that performance remains stable across a broad range of target leak values rather than depending on one exact optimum. We therefore present the paper as an empirical study of controlled auxiliary leak paths: the central claim is not universal superiority, but that leak can be formalized, learned, and stress-tested as a small auxiliary computation channel.

1 Introduction

The loose-screw project starts from a simple premise: leak in an activation function need not be treated only as a flaw. If the leak is bounded, controlled, and learnable, it can be studied as a small auxiliary signal path rather than as a defect to remove.

The main question in this paper is therefore narrow: can a model use a small leak channel as a stable helper without losing the benefits of a clean primary path? The main formulation studied here, Mixed Dual-Path Rattle, is built around that question. Earlier variants such as Pressure Rattle and Dual-Path Rattle are retained because they explain how the framework developed, but they are not the main result.

The framework evolves through three steps:

1. a pressure-aware negative-branch prototype,
2. a dual-path formulation with separate clean and leak channels, and
3. a mixed formulation in which the model learns how much each path should contribute.

The contribution is not any single primitive in isolation. ReLU, sigmoid gating, hyperbolic tangent saturation, and learned weights are all familiar components. The contribution is the way they are arranged into a leak-aware activation family whose behavior can be measured across real datasets, depth settings, noise levels, and leak-band sweeps.

More specifically, the paper makes four contributions:

- it defines a leak-aware activation family that progresses from a pressure-aware prototype to a mixed dual-path formulation with learnable clean and leak contributions;

- it identifies the mixed dual-path activation as the main practical variant and gives a compact boundedness interpretation for its components;
- it evaluates the family across multiple real datasets, depth settings, noise conditions, and ablation modes, with attention to dead-unit behavior and learned path usage; and
- it argues from the resulting sweeps that the family is better described by a task-dependent leak band than by a single universally optimal leak value.

2 Related Work

The proposed family is related to several established ideas. ReLU introduced sparse activations and simple gradient flow, while Leaky ReLU and PReLU address dead units by keeping a nonzero negative slope. Gated activations and mixture-based nonlinearities show that learned routing can improve representational flexibility. Residual pathways and auxiliary branches similarly support the idea that a network can benefit from multiple signal routes rather than a single rigid transformation.

The present framework is different in emphasis rather than in any single primitive. It does not claim novelty for sparsity, gating, bounded saturation, or learned mixing in isolation. Instead, it treats leak as a bounded auxiliary channel with a learnable contribution, and studies how that channel behaves under unconstrained training and under forced leak-band sweeps. The empirical question is therefore not whether leak exists, but how a controlled leak path behaves as a design choice.

3 Main Method: Mixed Dual-Path Rattle

The main formulation used throughout the paper is the **Mixed Dual-Path Rattle Activation**. For an input x , define a clean path and a bounded leak path as follows:

$$\begin{aligned} \text{clean}(x) &= \text{ReLU}(x), & p &= |x|, & \alpha(p) &= \sigma(ap + b), & \lambda &= \sigma(\ell), \\ \text{leak}(x) &= \lambda \alpha(p) \tanh(x). \end{aligned}$$

Let the learnable path weights be obtained from logits (θ_c, θ_l) through

$$[w_c, w_l] = \text{softmax}([\theta_c, \theta_l]).$$

The final activation is then

$$f(x) = w_c \text{clean}(x) + w_l \text{leak}(x).$$

This formulation preserves a standard primary signal path while giving the model a second bounded channel whose contribution can be learned rather than fixed by hand.

3.1 Why the design is stable

The stability of the framework comes from three simple components. The ReLU clean path preserves a strong primary gradient route on positive activations. The sigmoid gate keeps the pressure response bounded between 0 and 1. The hyperbolic tangent term keeps the auxiliary path smooth, sign-aware, and bounded, preventing it from exploding at large magnitude. In the mixed formulation, the softmax path weights further constrain the clean and leak contributions to remain normalized and nonnegative.

3.2 Interpretation

The model does not treat leak as the dominant signal. Instead, leak is used as a small, learned helper path. That is the interpretation that emerges repeatedly in the experiments: the auxiliary channel is not deleted, but it usually remains small.

4 Main Results

The main benchmark combines four real datasets: Wisconsin Breast Cancer, Wine, Digits, and Diabetes. Across three random seeds, the mixed formulation is competitive in several classification settings, especially where dead units matter, but it is not uniformly best. Its strongest results are not raw accuracy wins; they are stable training behavior, very low dead-unit rates, and consistent evidence that useful leak values occupy a task-dependent band rather than a single universal optimum.

The most important pattern is therefore not that Mixed Dual-Path Rattle beats every baseline everywhere. It is that the model usually preserves activity, remains trainable, and learns a small auxiliary contribution rather than collapsing the leak path to zero. On classification tasks, the mixed model is often close to the strongest baseline. On regression and deeper settings, the advantage is weaker and sometimes absent.

4.1 Dead units and stability

One of the clearest results is the dead-unit comparison. ReLU shows noticeable dead-unit rates that increase with depth, especially on Breast Cancer, Wine, Digits, and Diabetes. Leaky ReLU maintains essentially zero dead units throughout. Mixed Dual-Path Rattle behaves much closer to Leaky ReLU than to ReLU: on most runs its dead-unit rate is exactly zero, and even when nonzero it remains small relative to ReLU.

The important point is that the mixed formulation appears to preserve the activity-protection benefit of having a nonzero auxiliary path, even when that path receives only a small learned weight.

4.2 Robustness to input noise

Under additive test-time input noise, Mixed Dual-Path Rattle is sometimes competitive but not uniformly more robust. On Breast Cancer, mean noisy accuracy is slightly better than Leaky ReLU at moderate noise and remains close to the baselines at larger noise. On Wine and Digits, the mixed model often trails Leaky ReLU once noise is added. On Diabetes, all methods change only slightly under the tested noise levels, and Mixed Dual-Path Rattle remains worse than Leaky ReLU.

So the present evidence supports a narrower claim: the mixed formulation can remain stable under noise, but the current implementation does not establish a general noise-robustness advantage over simpler baselines.

4.3 Learned path mixtures

The learned mixture weights are especially informative. Across all four datasets and all tested depths, the mixed model consistently assigns the vast majority of weight to the clean path and only a small but nonzero weight to the leak path. The clean-path weight is typically around 0.95–0.96 on the classification datasets and about the same on Diabetes, leaving only a few percent for the leak path.

This is important because it gives a concrete interpretation of the framework: in the unconstrained mixed model, the leak path is used consistently, but sparingly. Validation loss decreases steadily while the learned mixture remains nearly flat, with the clean weight staying near 0.95 and the leak weight near 0.04–0.05 throughout training. Meanwhile, λ stays close to 0.10 and the average gate value α rises gradually over training, indicating that the model mainly adjusts the gate statistics while preserving a strongly clean-dominant mixture.

At the same time, the broader forced-leak experiments show that this natural small-leak preference should not be interpreted as a universal optimum. When the leak level is fixed externally and swept over a wide range, performance often remains competitive well beyond the natural 4–5% setting. The more defensible interpretation is therefore two-part: unconstrained optimization tends to settle on a small auxiliary leak, but the family as a whole appears to operate within a broader task-dependent leak band rather than around a single magic value.

4.4 Task summary

The overall pattern is cautious rather than dramatic. Mixed Dual-Path Rattle is viable and often competitive, but it is not uniformly best. On Breast Cancer it essentially ties the strongest baselines while almost eliminating dead units. On Digits it is competitive at shallow depth but loses ground at depth 6. On Wine it matches the baselines at depth 2 but degrades more strongly with additional depth. On Diabetes regression it underperforms Leaky ReLU across all depths, although it still preserves very low dead-unit rates.

These numbers suggest a modest but fairly consistent picture. The mixed formulation looks most promising on shallow-to-medium-depth classification settings, but it is not yet a reliable drop-in

replacement for standard activations across all tasks. Its clearest empirical profile is competitive behavior in some regimes, very low dead-unit rates, and evidence that useful leak values occupy a task-dependent band rather than a single universal optimum. Read conservatively, the benchmark supports viability and interpretability more strongly than it supports raw across-the-board gains.

4.5 Breast Cancer ablation: natural vs. forced leak weight

To test whether the small learned leak weight is merely an optimization accident or a meaningful preference, we ran an additional ablation on the real Wisconsin Breast Cancer classification dataset using a two-hidden-layer multilayer perceptron with Mixed Dual-Path Rattle activations. The baseline version used the standard softmax mixture,

$$[w_c, w_l] = \text{softmax}([\theta_c, \theta_l]),$$

while the modified version explicitly prevented the leak path from becoming too small by imposing a minimum leak contribution of 0.2. In code, this changes the mixture from a free softmax allocation to a constrained form in which

$$w_c = (1 - \delta) \tilde{w}_c, \quad w_l = (1 - \delta) \tilde{w}_l + \delta,$$

with $\delta = 0.2$ and $[\tilde{w}_c, \tilde{w}_l] = \text{softmax}([\theta_c, \theta_l])$.

The unconstrained model again learned a strongly clean-dominant allocation, with mean mixture approximately $w_c \approx 0.955$ and $w_l \approx 0.045$ by the end of training. Its final test accuracy was about 0.9649, and the `clean_only` evaluation matched that same accuracy, while the `leak_only` path remained much weaker at about 0.6053.

When the leak path was forced to keep at least 20% of the mixture mass, the learned allocation shifted to approximately $w_c \approx 0.764$ and $w_l \approx 0.236$. Performance dropped: the full mixed model reached only about 0.9035 test accuracy, while `clean_only` still performed better at about 0.9123. The `leak_only` result again stayed near 0.6053.

This ablation is informative because it suggests that the small leak weight is not merely a cosmetic artifact of initialization in this setting. On this real tabular classification problem, the model appears to prefer using the leak path sparingly. Forcing the leak contribution upward weakens performance rather than strengthening it. That behavior is consistent with the framework interpretation developed above: the leak path is useful primarily as a bounded auxiliary correction, not as a co-equal replacement for the clean path.

4.6 Empirical mapping of the leak band

To better understand the role of the auxiliary leak channel, we performed a focused sweep over fixed target leak values ranging from 0.00 to 0.50 on depth-4 multilayer perceptrons for Digits and Breast Cancer.

The results reveal a consistent pattern across the two datasets. Rather than exhibiting a sharp optimum, performance remains relatively stable across a wide range of leak values. On Digits,

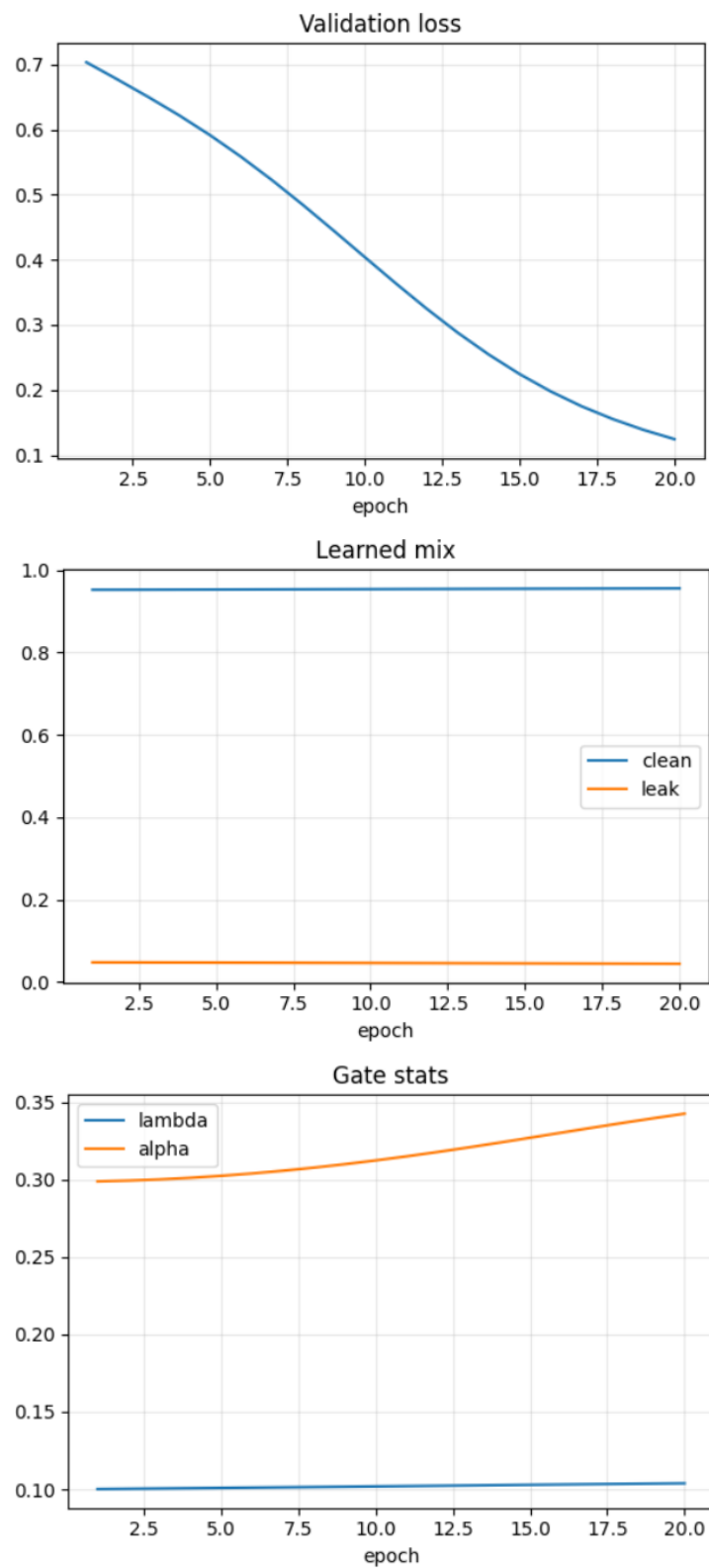


Figure 1: Mixed-model training trajectory on the Breast Cancer dataset. Validation loss falls steadily, the clean-path weight stays near 0.95, the leak-path weight stays near 0.05, and the gate statistics change only gradually.

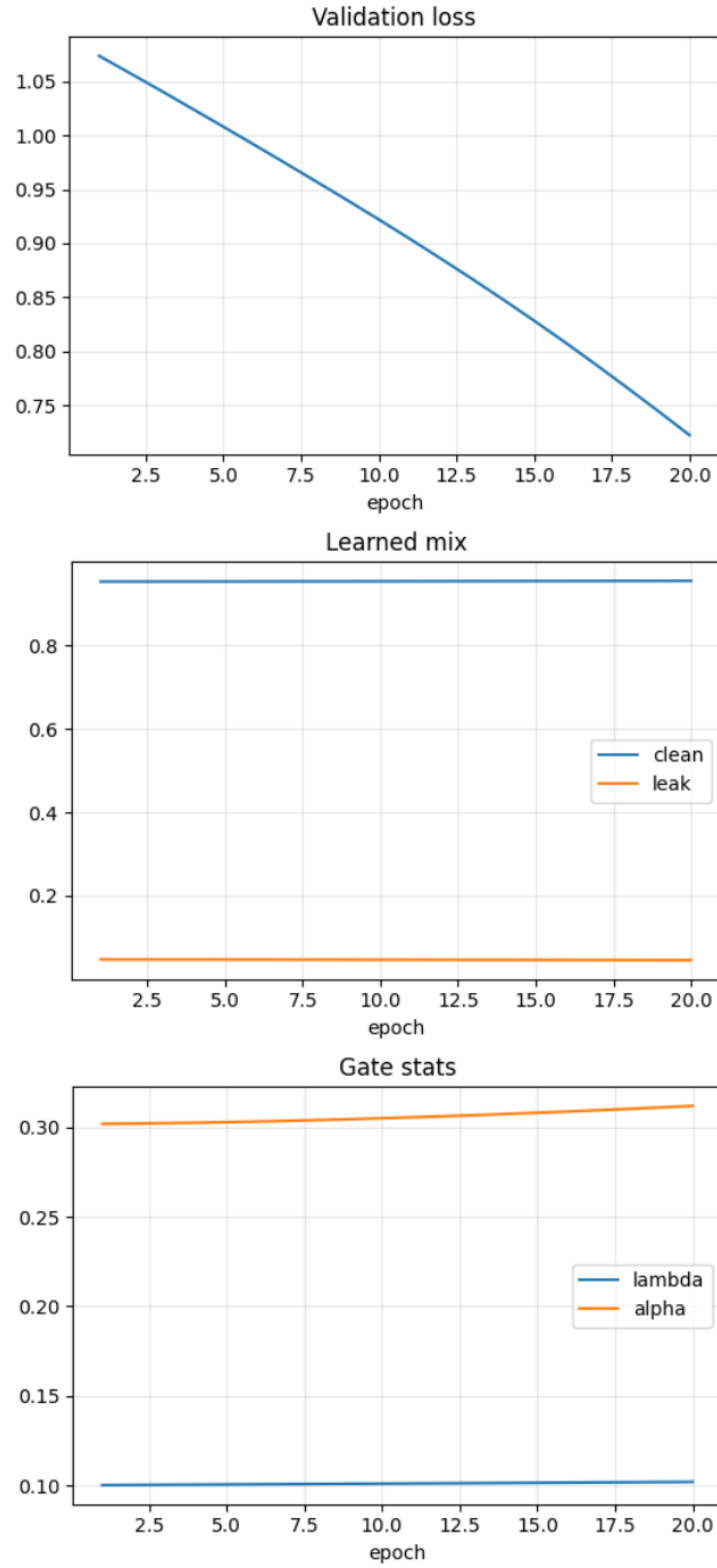


Figure 2: Mixed-model training trajectory on the Wine dataset. The same pattern appears: decreasing validation loss, nearly flat clean/leak mixture weights, and slow gate adaptation.

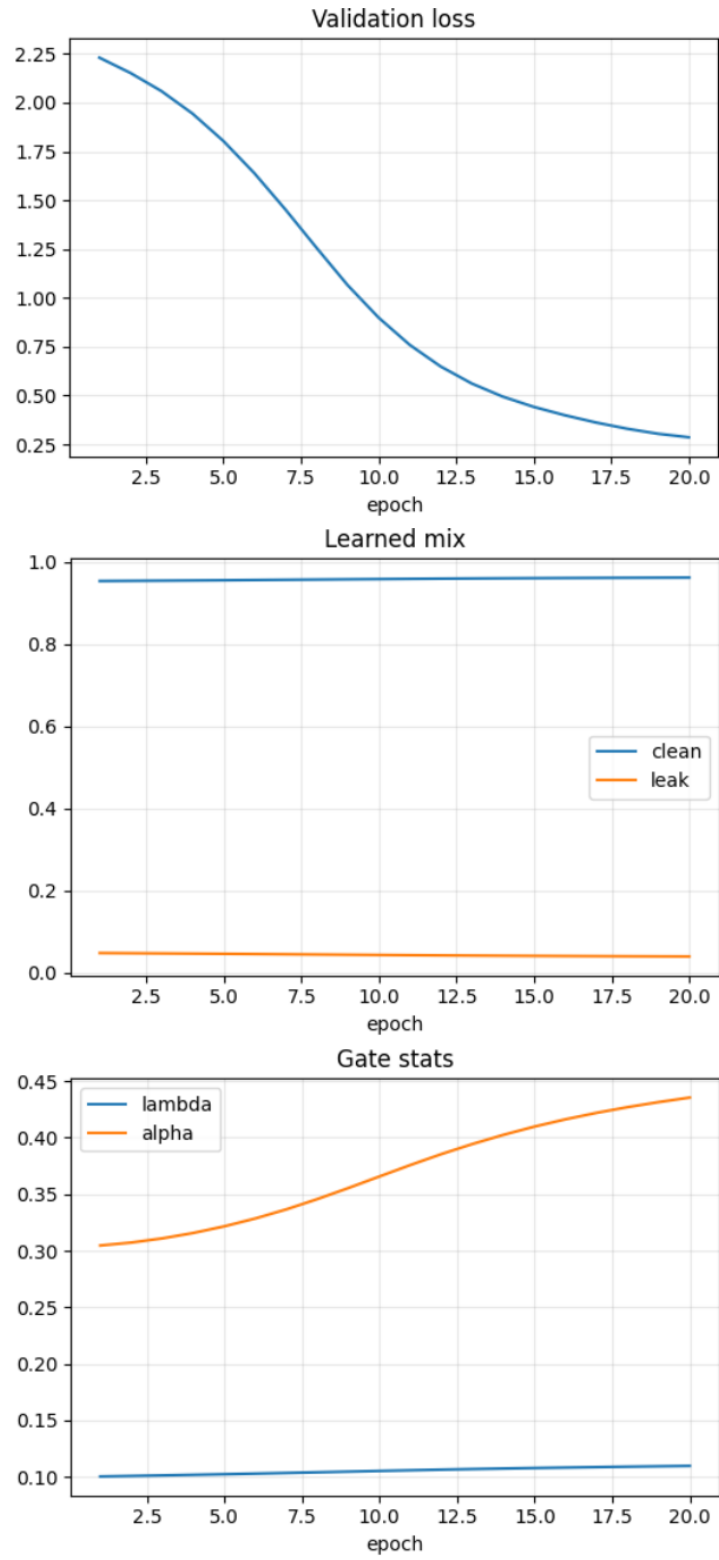


Figure 3: Mixed-model training trajectory on the Digits dataset. The clean path remains dominant throughout training while the gate value α increases gradually.

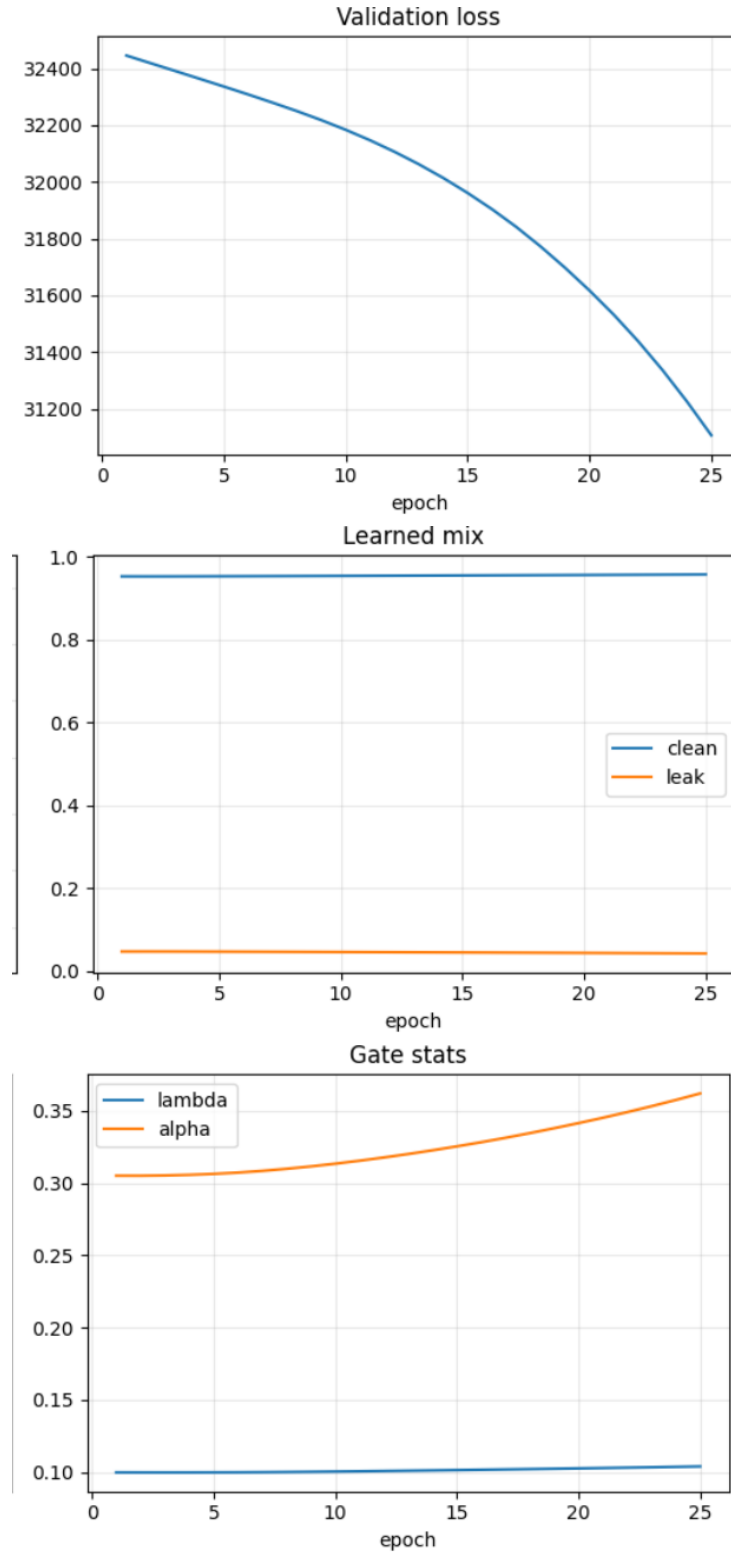


Figure 4: Mixed-model training trajectory on the Diabetes dataset. The learned mixture again stays almost fixed, even though the regression setting is empirically weaker than the classification settings.

mean test accuracy stays within a narrow band across all tested leak levels, with the strongest averages appearing at very small and moderately large leak values. On Breast Cancer, mean test accuracy stays between roughly 0.96 and 0.98, with the strongest averages occurring at moderate leak values.

These findings suggest that the framework does not depend on a precise universal leak value. Instead, it appears to operate within a broad *leak band* in which the auxiliary path can be varied without strongly affecting performance. This is more informative than a single optimum: it suggests that the leak channel behaves as a robust structural component rather than as a fragile parameter that must be tuned to one narrow setting.

Importantly, even relatively large leak values such as 0.50 do not cause performance collapse in this focused sweep. That behavior is consistent with the bounded leak design helping prevent the auxiliary path from overwhelming the primary signal, even when the imposed leak level is much larger than the naturally learned mixture weight in the unconstrained model.

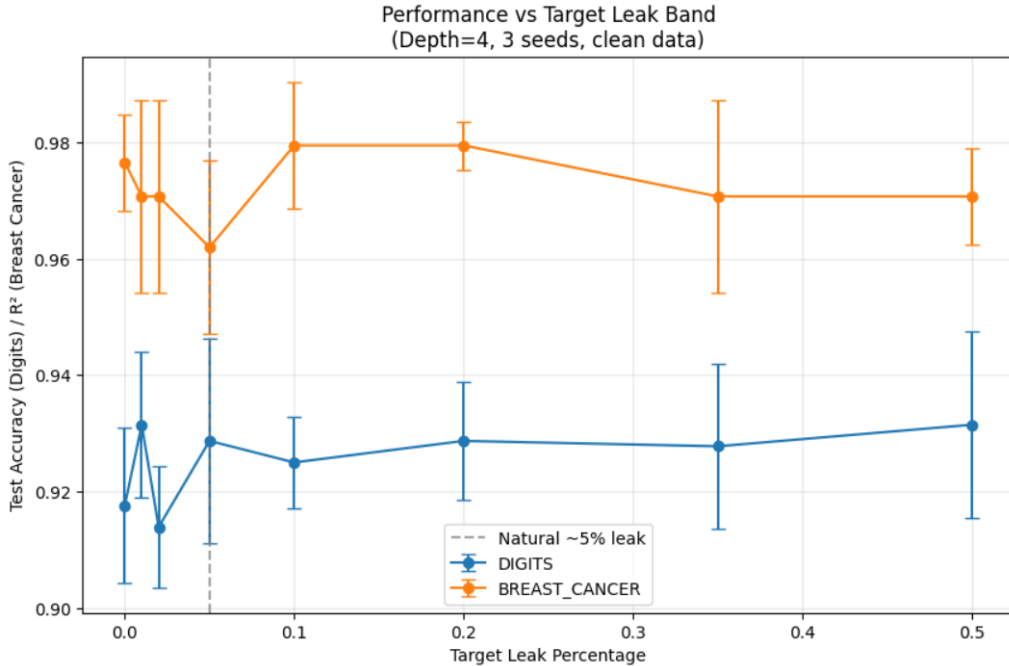


Figure 5: Focused leak-band sweep on Digits and Breast Cancer with depth-4 multilayer perceptrons over three seeds. Performance remains strong across a broad range of target leak values, supporting the interpretation of a task-dependent leak band rather than a single privileged optimum. The dashed line marks the natural small-leak regime near 5%.

4.7 Target-leak sweep across real datasets

We then ran a broader additive-leak benchmark across Breast Cancer, Wine, Digits, and Diabetes. In this setting the activation again takes the form

$$\text{clean}(x) + w_l \text{leak}(x),$$

but now the target leak was swept over 0.05, 0.20, 0.35, and 0.50, while the penalty strength was swept over 0.0, 0.1, 0.5, and 1.0. This lets us separate two questions: whether the framework benefits from different leak bands, and whether stronger penalties substantially change the learned leak within a chosen band.

The clearest result is that the penalty strength mattered much less than the target leak itself. Within a fixed target, the learned leak means stayed extremely close to that target across seeds and datasets. For example, Breast Cancer settled near the target values across the sweep; Digits behaved similarly; and comparable behavior appeared on Wine and Diabetes. So in this benchmark the penalty mainly holds the system inside a chosen leak band rather than creating strong additional adaptation by itself.

The task-level behavior is also informative. On Breast Cancer and Wine, the strongest forced configurations were shallow models with very small target leak, typically around 0.05, and they matched or slightly exceeded the standard baselines while keeping dead-unit rates at zero. On Digits, larger target leaks were often better: for the full dataset at depth 4, the best mean forced result was above both ReLU and Leaky ReLU, and on the smaller setting at depth 8 the best forced results were around the upper end of the tested band. On Diabetes regression, the picture was again different: moderate or small target leaks performed best, and the strongest forced configurations were competitive with the standard baselines.

The noise sweeps reinforce the same idea. Higher evaluation noise did not fundamentally reorder the preferred leak band within each task. Instead, configurations that were already strong at zero noise usually remained strong under noise, which suggests that the target leak is acting more like a structural bias on representation than like a separate robustness knob that can be tuned independently through penalty strength.

The main lesson from these benchmarks is that the loose-screw family does not appear to have a single universal best leak value. Rather, different tasks prefer different leak bands: very small leak for some tabular classification settings, moderate leak for some Breast Cancer settings, and larger leak values can remain viable on Digits without causing collapse. This strengthens the framework interpretation considerably. The useful question is no longer simply whether leak should be near zero, but which controlled leak band is most appropriate for a given regime.

5 Interpretation and Scope

The central empirical finding of this study is that the leak-aware family exhibits a broad, task-dependent operating region rather than a single privileged leak value. In unconstrained mixed models, optimization naturally settles on a small auxiliary leak of roughly 4–5%. But when the leak is fixed externally and swept across a much wider range, performance often remains strong over a broad interval instead of collapsing away from that natural equilibrium. This makes the main significance of the work more specific than a broad performance claim. Its value is that it identifies and formalizes a design pattern in which leak is treated as a bounded auxiliary signal path rather than only as an error term.

The strongest current claim should therefore be limited and defensible. The mathematical ingre-

dients are familiar: ReLU, sigmoid gating, hyperbolic tangent saturation, additive side paths, and softmax mixing. What is new here is their arrangement into a leak-aware activation family in which pressure-aware leak is explicitly modeled, separated, and then mixed with the primary path in a learnable way.

A defensible description is the following: this is a **novel activation-family direction** in which leak is elevated from a passive residual effect to a controlled auxiliary channel, and in which the mixed and additive formulations let the network regulate how much of that auxiliary channel should be used. Empirically, the clearest current result is not universal superiority, but the combination of two behaviors: a natural tendency toward small learned leak under unconstrained mixing, and a broader leak band within which forced leak remains viable. The supplementary deep-CNN and MNIST follow-ups fit that interpretation as diagnostic evidence rather than as headline wins: they mainly indicate that once optimization is stabilized, especially by normalization, the mixed formulation remains viable in harder regimes without clearly surpassing the strongest baseline.

This avoids overclaiming. It does not say that every component is individually unprecedented, nor does it claim universal superiority over standard activations. Instead, it claims that the family perspective, the progression of variants, and the observed leak-band behavior together constitute a real and testable contribution.

6 Limitations

The current evidence is useful but not complete. The benchmarks are still limited in scope, and the strongest results are not consistently better than strong baselines. The mixed formulation is best viewed as an interpretable and empirically testable family rather than as a universal replacement for standard activations.

Three limitations matter most. First, the evaluation is broad enough to reveal patterns, but still too small to support claims about modern large-scale training regimes. Second, the paper compares against strong baseline activations but not against a wider set of modern gated or adaptive alternatives. Third, several of the most interesting observations—especially the broader leak-band behavior and the deep-network diagnostics—are empirical regularities rather than consequences of a formal theory.

The current results also suggest that a better empirical question is not simply whether leak helps, but when and how much leak helps. That is why the leak-band view is central: the family appears to admit a useful operating region rather than a single best value.

7 Next Experimental Direction

The next research question is no longer whether the idea can be written down mathematically. It is whether the leak-aware family has a useful operating region and whether that region remains stable in harder settings.

The most useful next steps are:

- sweep the target leak level directly, from very small to moderately large values, to test whether the framework has a useful leak band rather than a single universal optimum;
- compare fixed, learnable, and per-neuron leak settings;
- test deeper networks and harder datasets, especially noisy-label and small-data regimes;
- track the learned values of a , b , λ , and the path weights throughout training; and
- extend the framework beyond activations into leak-aware blocks or lightweight residual-style modules.

This progression keeps the work grounded: formalize the variants that were actually run, report what the benchmark really shows, and use the current results as a base for more targeted experiments.

8 Conclusion

The current paper does not establish a universally superior activation. Instead, it establishes a more limited but still meaningful result: leak can be treated as a bounded auxiliary signal path, and the resulting family behaves in a controlled and task-dependent way.

The strongest current evidence is that the mixed dual-path formulation naturally learns a small leak contribution, while forced leak-band sweeps show that performance can remain stable across a broad range of leak values. The supplementary deep-network diagnostics point in the same direction but should be read more cautiously: they mainly show that the formulation remains analyzable in harder regimes, and that optimization choices such as normalization can dominate outcomes there.

Taken together, these results support treating the loose-screw family not as a single magical activation, but as a structured framework for studying how small auxiliary signal channels behave in neural networks. The main takeaway is therefore conceptual as much as empirical: a leak channel can be present, bounded, and useful without becoming the dominant path, and that controlled auxiliary role is measurable across both unconstrained training and forced leak sweeps.

A Appendix A: Earlier activation variants

This appendix collects the earlier designs that motivated the main mixed dual-path formulation.

A.1 Pressure Rattle Activation

The implementation that produced the reported sample outputs is the **Pressure Rattle Activation**. For an input x , define local pressure as

$$p = |x|.$$

The looseness gate is

$$\alpha(p) = \sigma(ap + b),$$

where $\sigma(\cdot)$ is the sigmoid function and a, b are learnable scalar parameters. A second bounded learnable factor is

$$\lambda = \sigma(\ell),$$

where ℓ is a learnable scalar logit. Using this, define the rattle factor

$$r(x) = \lambda \alpha(p) \tanh(p).$$

The activation is then piecewise defined by

$$f(x) = \begin{cases} \max(0, x) = x, & x \geq 0, \\ x(1 - r(x)), & x < 0. \end{cases}$$

This version keeps the positive branch unchanged and contracts the negative side toward zero by a pressure-aware bounded amount.

A.2 Universal Loose Screw family

A second code block explored a broader family called **Universal Loose Screw**, with three modes: **boring**, **safe**, and **rattle**. In that family, one first defines

$$p = |x|, \quad \alpha(p) = \sigma(sp),$$

where s is a learnable pressure sensitivity, and introduces a learnable looseness parameter L .

The three negative-branch variants are then:

$$\begin{aligned} \text{boring: } n(x) &= 0.01x, \\ \text{safe: } n(x) &= x(1 - L\alpha(p) \tanh(p)), \\ \text{rattle: } n(x) &= x + L\alpha(p) \tanh(x^2). \end{aligned}$$

The full activation in each case is

$$f(x) = \text{ReLU}(x) + \mathbf{1}_{\{x < 0\}} n(x).$$

Among these, the **safe** mode is the closest conceptual match to the Pressure Rattle Activation above, because it keeps the negative-side correction bounded and visually stable. The **rattle** mode is more exploratory and less clean as a first formal definition.

A.3 Dual-Path Rattle Activation

A later benchmark introduced a different formulation, **Dual-Path Rattle Activation**, that returns to the idea of a separate auxiliary channel rather than only deforming the negative branch. For an input x , define the clean path by

$$\text{clean}(x) = \text{ReLU}(x).$$

Let pressure and the pressure-aware gate be

$$p = |x|, \quad \alpha(p) = \sigma(ap + b),$$

and again define a bounded learnable leak scale

$$\lambda = \sigma(\ell).$$

The separate bounded leak path is then

$$\text{leak}(x) = \lambda \alpha(p) \tanh(x).$$

The activation output is

$$f(x) = \text{clean}(x) + \text{leak}(x).$$

This form differs from Pressure Rattle Activation in one important respect: it preserves the clean ReLU path and adds a second bounded signal channel.

B Appendix B: Forced leak ablation details

To test whether the small learned leak weight is merely an optimization accident or a meaningful preference, we ran an additional ablation on the real Wisconsin Breast Cancer classification dataset using a two-hidden-layer multilayer perceptron with Mixed Dual-Path Rattle activations. The baseline version used the standard softmax mixture,

$$[w_c, w_l] = \text{softmax}([\theta_c, \theta_l]),$$

while the modified version explicitly prevented the leak path from becoming too small by imposing a minimum leak contribution of 0.2.

In code, this changes the mixture from a free softmax allocation to a constrained form in which

$$w_c = (1 - \delta) \tilde{w}_c, \quad w_l = (1 - \delta) \tilde{w}_l + \delta,$$

with $\delta = 0.2$ and $[\tilde{w}_c, \tilde{w}_l] = \text{softmax}([\theta_c, \theta_l])$.

The unconstrained model learned a clean-dominant allocation, with mean mixture approximately $w_c \approx 0.955$ and $w_l \approx 0.045$. Its final test accuracy was about 0.9649, and the `clean_only` evaluation matched that same accuracy, while the `leak_only` path remained much weaker at about 0.6053.

When the leak path was forced to keep at least 20% of the mixture mass, the learned allocation shifted to approximately $w_c \approx 0.764$ and $w_l \approx 0.236$. Performance dropped: the full mixed model reached only about 0.9035 test accuracy, while `clean_only` still performed better at about 0.9123. The `leak_only` result again stayed near 0.6053.

C Appendix C: Target-leak sweep across real datasets

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The clearest result is that penalty strength mattered much less than target leak itself. Within a fixed target, the learned leak means stayed extremely close to that target across seeds and datasets. For example, Breast Cancer settled near the target values across the sweep; Digits behaved similarly; and comparable behavior appeared on Wine and Diabetes.

The task-level behavior was also informative. On Breast Cancer and Wine, the strongest forced configurations were shallow models with very small target leak, typically around 0.05, and they matched or slightly exceeded the standard baselines while keeping dead-unit rates at zero. On Digits, larger target leaks were often better. On Diabetes regression, moderate or small target leaks performed best. The noise sweeps reinforced the same idea: higher evaluation noise did not fundamentally reorder the preferred leak band within each task.

D Appendix D: Supplementary CIFAR-10 and 20 Newsgroups stress tests

This appendix collects supplementary stress-test results from the image and text benchmarks. These runs provide additional context and boundary-condition diagnostics, but they should be read as supplementary rather than as the paper’s central evidence.

In the CIFAR-10 depth sweep, the deep CNN benchmark showed that Leaky ReLU was slightly ahead on average, while the mixed formulation remained competitive and preserved zero dead units. In the leak-band sweep, performance stayed in a relatively narrow range from target leak 0.00 to 0.50, and the learned or fixed leak setting did not cause collapse. These image experiments are best read as supplementary diagnostics: they suggest that the auxiliary leak path can remain usable under a harder benchmark, but they do not show a consistent advantage over the strongest baseline.

In the 20 Newsgroups benchmark, the mixed model was again competitive with ReLU and Leaky ReLU, with small differences across depth and noise settings. The main value of this run is qualitative: it is consistent with the same broad pattern seen in the tabular experiments, namely that the

mixed formulation behaves like a usable auxiliary-path activation rather than an obviously fragile construction.

Table 1: Supplementary 20 Newsgroups summary from the stress-test benchmark. Values are three-seed means for the full-data setting.

Depth	Activation	Clean acc.	Noisy acc.	Dead rate
2	ReLU	0.6371	0.6179	0.0020
2	Leaky ReLU	0.6374	0.6176	0.0000
2	Mixed	0.6368	0.6183	0.0000
4	ReLU	0.6097	0.5911	0.0825
4	Leaky ReLU	0.6134	0.5944	0.0000
4	Mixed	0.6179	0.5943	0.0000

Table 1 makes the omitted text benchmark concrete. At depth 2, all three activations are essentially tied on clean accuracy, while at depth 4 the mixed model is marginally best on clean accuracy and remains tied with Leaky ReLU under the noisy condition. The dead-rate pattern is the same as in the other experiments: ReLU begins to lose active units with depth, whereas Leaky ReLU and the mixed formulation remain effectively at zero.

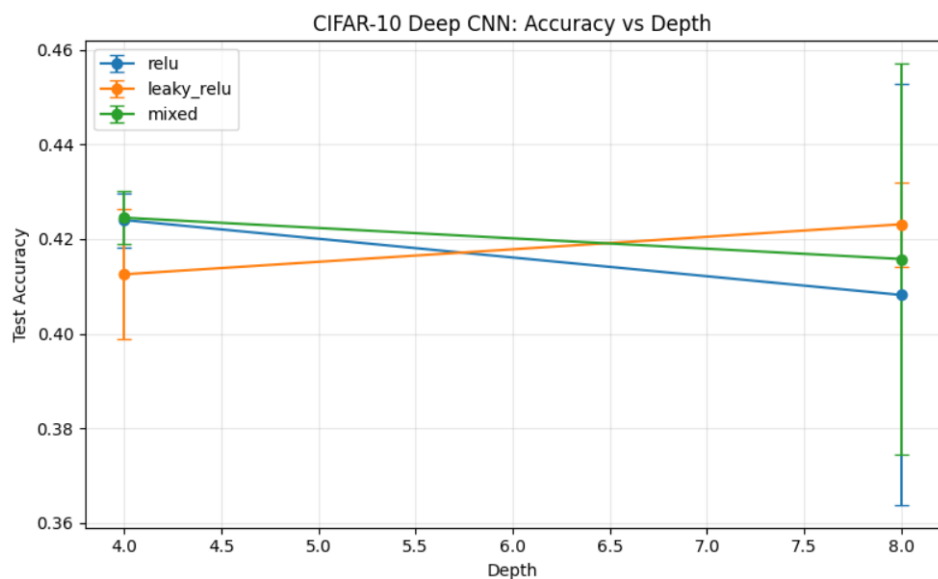


Figure 6: CIFAR-10 deep-CNN accuracy versus depth for ReLU, Leaky ReLU, and Mixed Dual-Path Rattle. The mixed model remains competitive but does not consistently beat the strongest baseline.

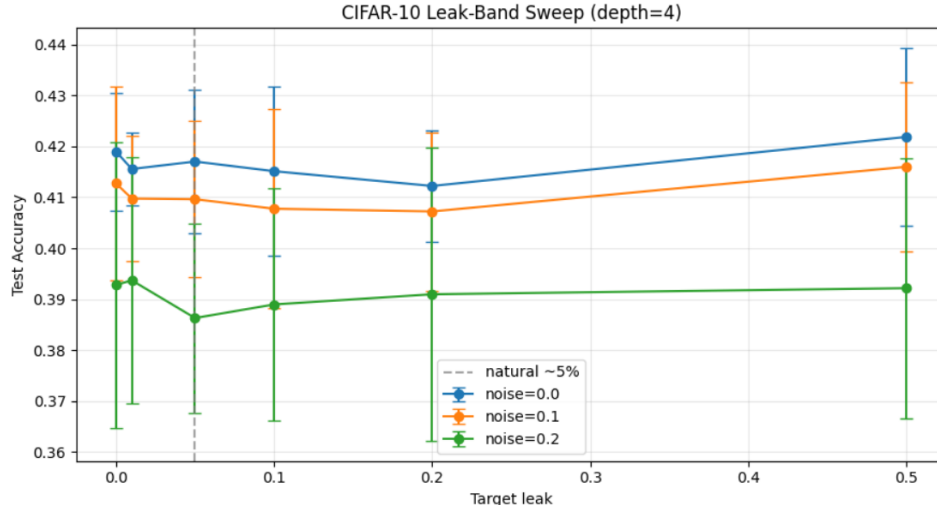


Figure 7: CIFAR-10 leak-band sweep at depth 4 under three evaluation noise levels. The forced leak weight tracks the target band without causing collapse, and test accuracy remains in a relatively narrow range from target leak 0.00 to 0.50.

D.1 Deep non-residual MLP stress test

We also ran a harder supplementary stress test on CIFAR-10 using a 25-layer non-residual multilayer perceptron with hidden width 128, trained for 5 epochs on a 50% stratified subset of the training set. This setting is intentionally harsh: it removes convolutional inductive bias, removes residual connections, and pushes depth high enough that dead-unit behavior becomes easier to detect.

Across three seeds, the mixed formulation achieved mean test accuracy about 0.4016 ± 0.0099 , compared with about 0.3858 ± 0.0076 for Leaky ReLU and about 0.3840 ± 0.0229 for ReLU. The dead-unit pattern was sharper: ReLU reached mean dead-unit rate about 0.1376, whereas both Leaky ReLU and the mixed formulation remained at 0.0000 under the same threshold. In the mixed model, the learned path weights again stayed strongly clean-dominant, with mean clean-path weight about 0.9455 and mean leak-path weight about 0.0545.

This result should still be interpreted cautiously. It does not show universal superiority, but it does strengthen one narrower part of the framework claim: in this very deep non-residual setting, the mixed auxiliary-path formulation remains trainable, preserves activity across all layers, and stays competitive with the standard baselines.

D.2 Breaking-point deep CNN stress test

We also ran a deeper supplementary breaking-point experiment on CIFAR-10 using plain convolutional networks without residual connections at depths 25 and 50. This test should be interpreted primarily as a failure-mode diagnostic rather than as a headline performance result. Its value is that it helps separate two questions: whether the mixed auxiliary-path activation remains stable

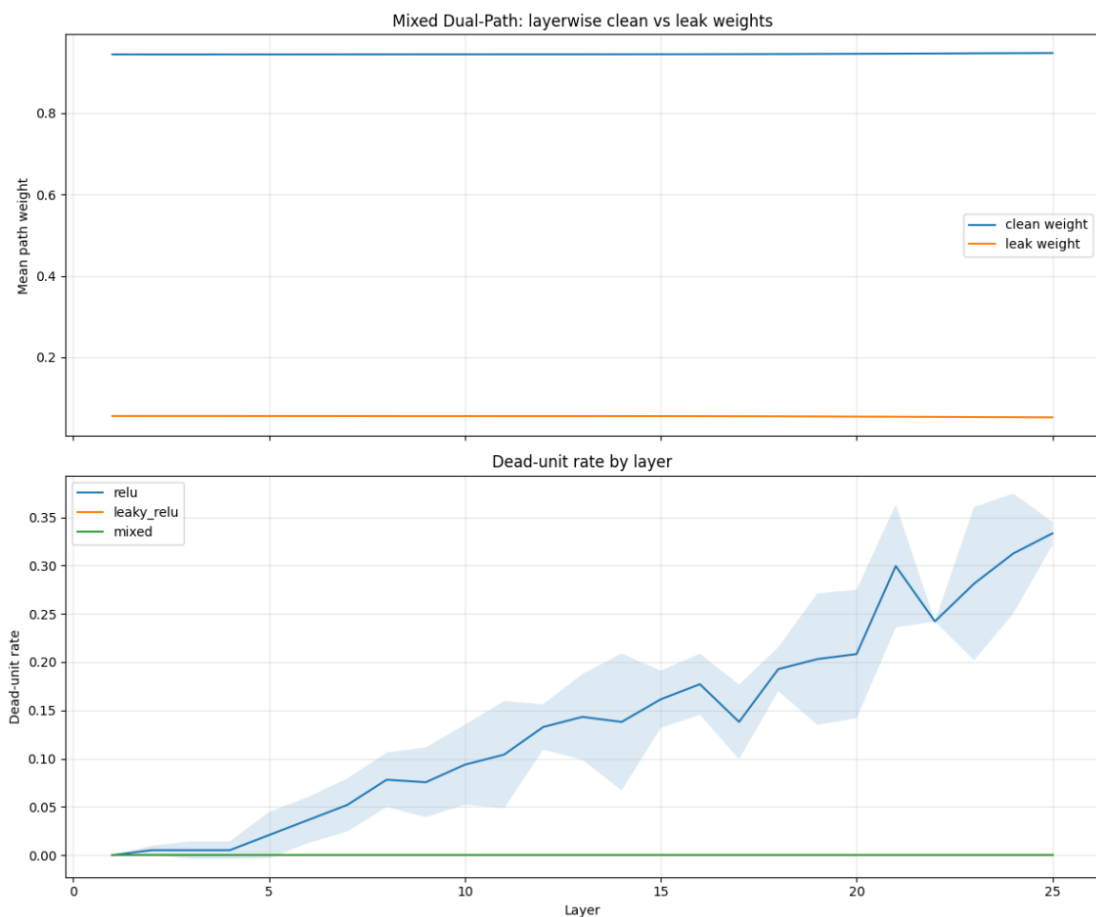


Figure 8: Layerwise summary from the 25-layer non-residual CIFAR-10 MLP stress test. The top panel shows that the mixed model keeps a strongly clean-dominant allocation with a small but persistent leak weight across depth. The bottom panel shows that ReLU accumulates dead units progressively with depth, while Leaky ReLU and the mixed formulation remain effectively at zero throughout the network.

in very deep settings, and whether that stability alone is sufficient to rescue an otherwise difficult architecture.

The empirical pattern is clear. At depth 25, Leaky ReLU was the strongest and most stable of the three activations, with mean test accuracy about 0.2213 ± 0.0300 , while the mixed model reached about 0.1705 ± 0.1221 and ReLU reached about 0.1537 ± 0.0930 . At depth 50, all three activations collapsed to chance-level accuracy near 0.10, indicating that the architecture itself had become the dominant bottleneck. In other words, this experiment does not show that the mixed formulation solves extreme depth by itself.

The diagnostic plots are still informative. ReLU and the mixed formulation accumulated large dead-unit rates as depth increased in this specific setup, whereas Leaky ReLU remained substantially lower. For the mixed model, the learned path allocation stayed almost entirely clean-dominant, with clean weight near 0.9975 and leak weight near 0.0025 across layers at both depths. That behavior is consistent with a very strongly suppressed auxiliary path in this run. It may reflect an initialization effect, but this experiment alone does not prove that the initialization is the sole cause of failure; a simpler and safer conclusion is that the very deep plain CNN is unstable for all three activations under the chosen training setup.

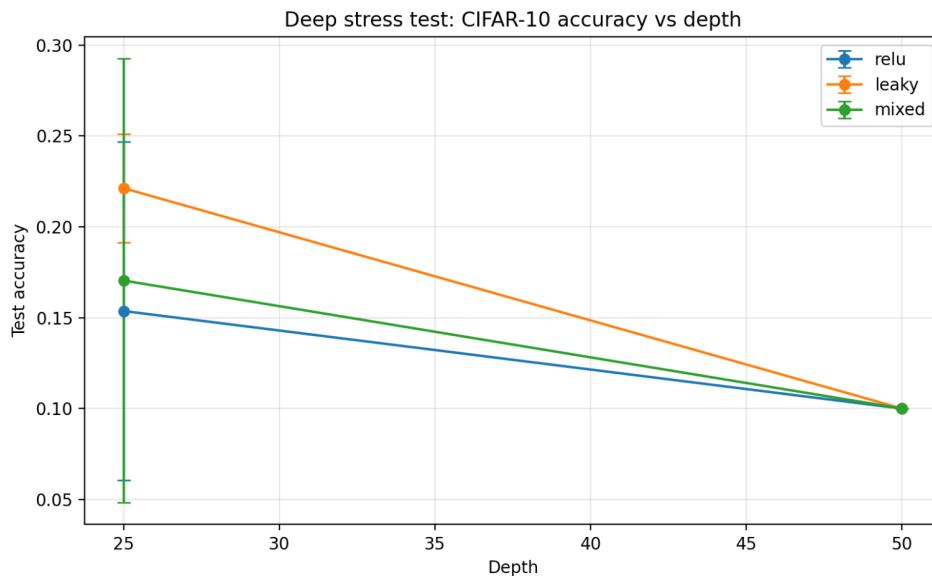


Figure 9: Breaking-point deep CNN stress test on CIFAR-10: test accuracy versus depth. Performance drops sharply from depth 25 to depth 50, where all three activations collapse to chance-level accuracy.

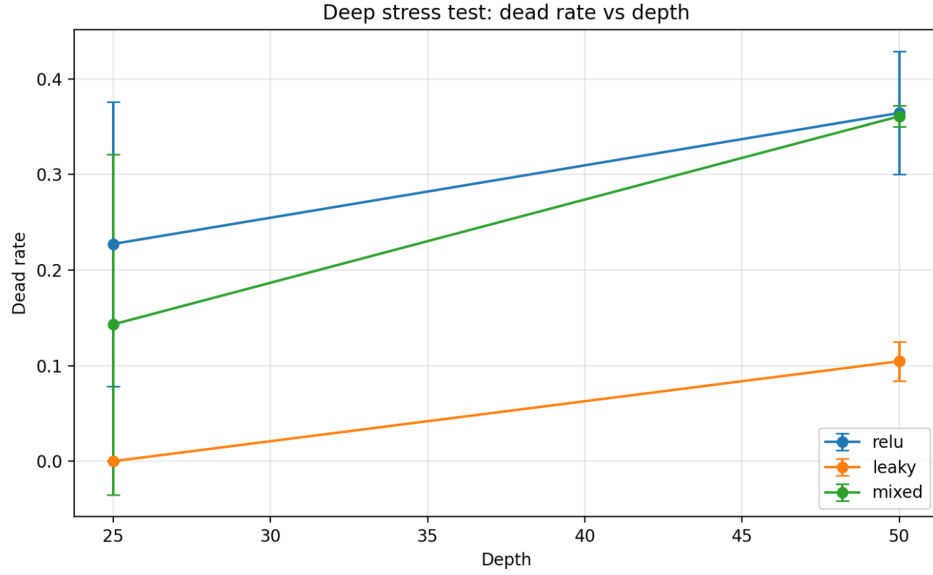


Figure 10: Breaking-point deep CNN stress test on CIFAR-10: dead-unit rate versus depth. ReLU and the mixed formulation accumulate substantial inactive units in this setup, while Leaky ReLU remains lower, though not zero at depth 50.

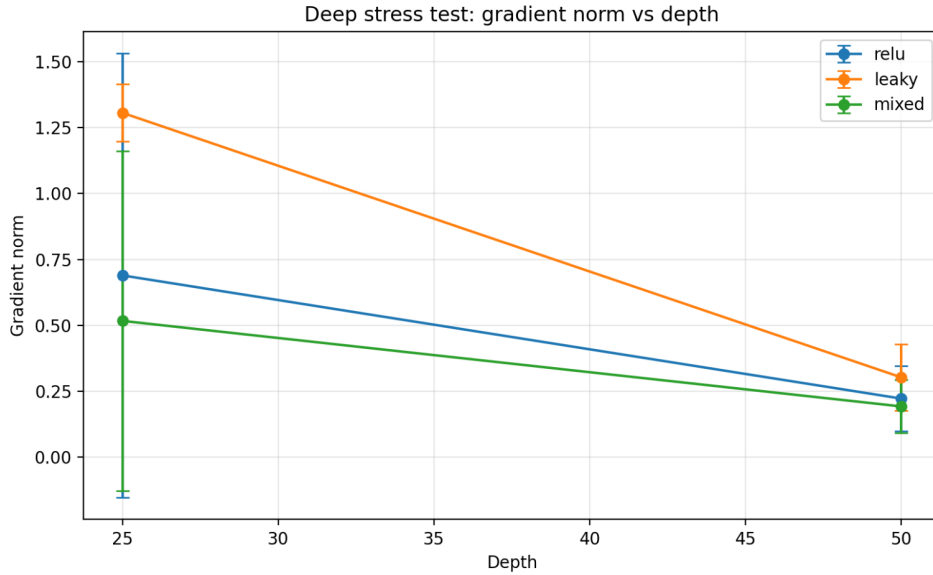


Figure 11: Breaking-point deep CNN stress test on CIFAR-10: gradient norm versus depth. The gradients shrink substantially by depth 50 for all three activations, supporting the interpretation that optimization difficulty is a major contributor to failure in this regime.

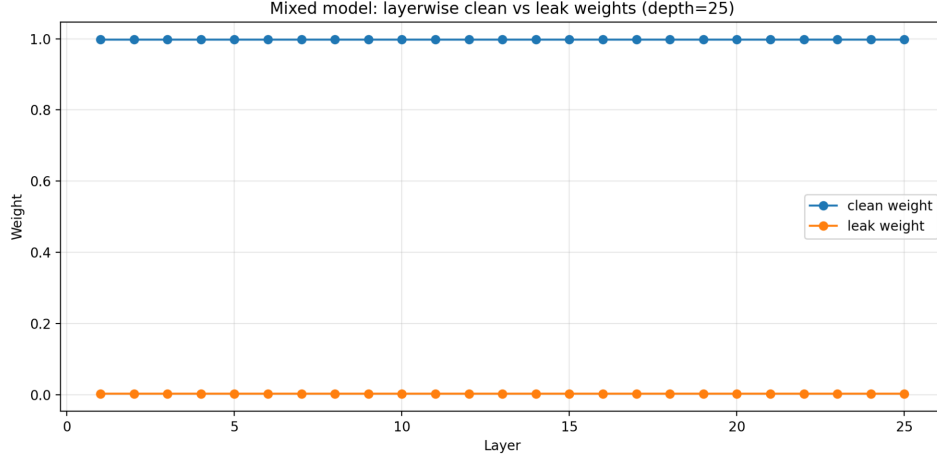


Figure 12: Layerwise clean versus leak weights for the mixed model at depth 25 in the breaking-point CNN experiment. The allocation remains overwhelmingly clean-dominant, with the leak path contributing only a very small fraction throughout the network.

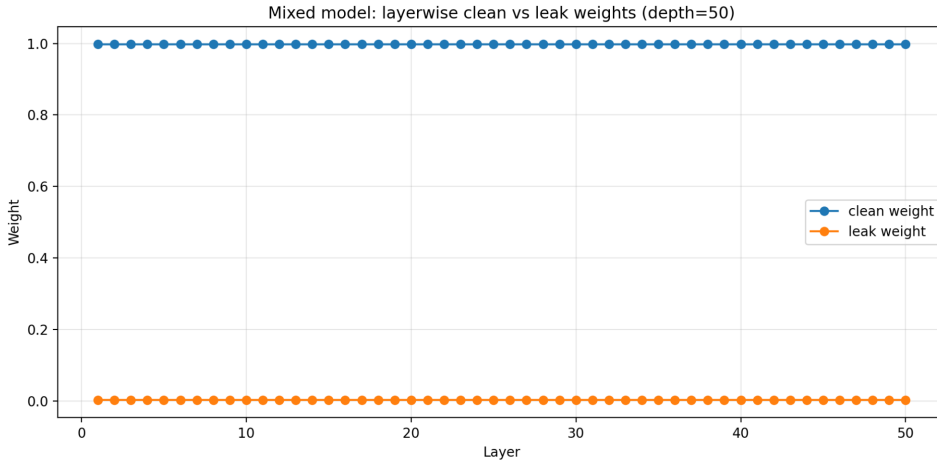


Figure 13: Layerwise clean versus leak weights for the mixed model at depth 50 in the breaking-point CNN experiment. The learned allocation is again almost entirely clean-dominant, indicating that the auxiliary path remains strongly suppressed in this failed-depth regime.

D.3 Depth-25 CNN follow-up: initialization versus normalization

To probe the breaking-point result more directly, we ran a focused follow-up at depth 25 on the same plain non-residual CNN, now comparing eight variants: ReLU, Leaky ReLU, and the mixed activation with either a roughly 95/5 clean-leak initialization or a 50/50 initialization, each with and without batch normalization. The question was narrower than in the earlier appendix experiment: is the poor depth-25 behavior mainly a consequence of the mixed-path initialization, or is normalization the larger factor?

The answer from this run is suggestive rather than definitive. Without batch normalization, all

four plain variants remained weak and unstable. Mean test accuracy was about 0.1801 ± 0.0717 for ReLU, 0.2164 ± 0.1065 for Leaky ReLU, 0.1416 ± 0.0721 for the mixed 95/5 model, and exactly chance-level 0.1000 for the mixed 50/50 model. Dead-unit rates were also substantial in the plain setting, especially for the 50/50 mixed initialization, which reached mean dead rate about 0.4000. So changing the initial clean-leak balance alone does not rescue the deep plain CNN; in fact, the symmetric initialization makes the no-batch-norm case worse.

With batch normalization, however, the picture changes materially. All four normalized variants trained reliably, all reached zero measured dead units under the same threshold, and their accuracies clustered in a much narrower band: about 0.4617 ± 0.0121 for ReLU, 0.4374 ± 0.0213 for Leaky ReLU, 0.4309 ± 0.0185 for mixed 95/5, and 0.4364 ± 0.0258 for mixed 50/50. This does not make the mixed formulation the best performer in the normalized regime, but it does suggest that the depth-25 failure mode is driven much more by optimization and normalization than by the auxiliary-path idea itself.

The learned mixtures add a useful qualification. In the no-batch-norm runs, the mixed 95/5 model stayed essentially fixed at its initial allocation, while the 50/50 model remained near parity and drifted slightly toward the leak path in later layers, yet still failed completely. In the batch-normalized runs, the mixed 95/5 model again stayed strongly clean-dominant, whereas the mixed 50/50 model learned a broader layerwise redistribution while still training successfully. That combination supports a narrower interpretation than a strong causal claim: batch normalization is the main stabilizer here, while the mixed activation remains viable once the surrounding optimization problem is made tractable.

The noise sweep is consistent with that reading. Over test-time Gaussian noise levels from 0.0 to 0.2, each variant changed only modestly relative to its own clean baseline. The normalized models remained stronger than the plain ones throughout, and there was no evidence that the mixed variants gained a unique robustness advantage in this particular follow-up.

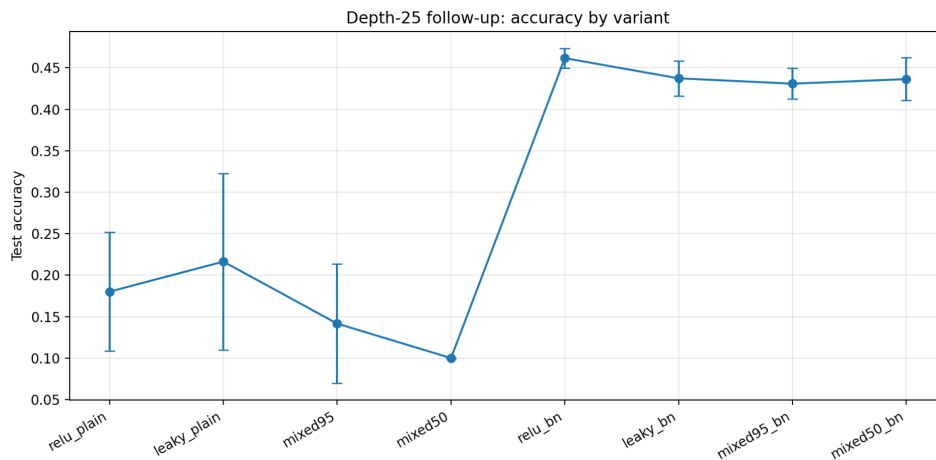


Figure 14: Depth-25 CIFAR-10 follow-up on the plain deep CNN: test accuracy across activation and normalization variants. The main separation is between no-batch-norm and batch-normalized models rather than between the mixed initializations themselves.

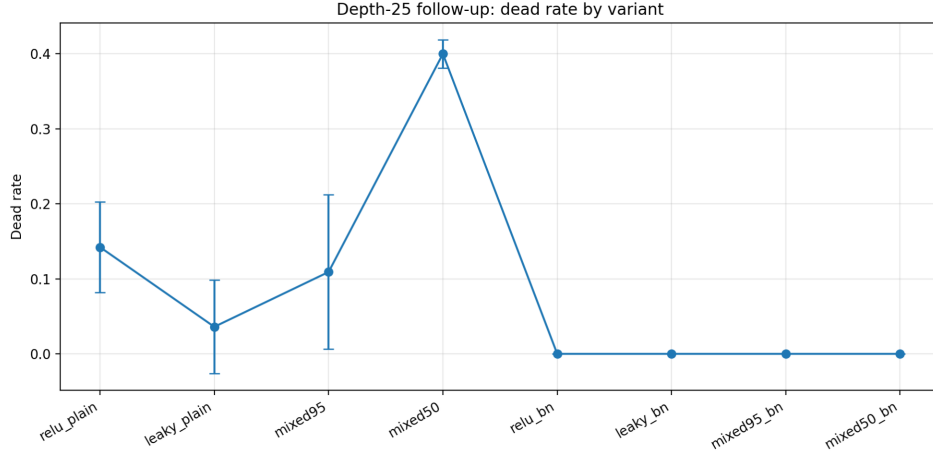


Figure 15: Depth-25 follow-up: mean dead-unit rate by variant. Batch normalization eliminates the dead-rate problem in this setting, while the plain 50/50 mixed initialization is the most fragile of the unnormalized models.

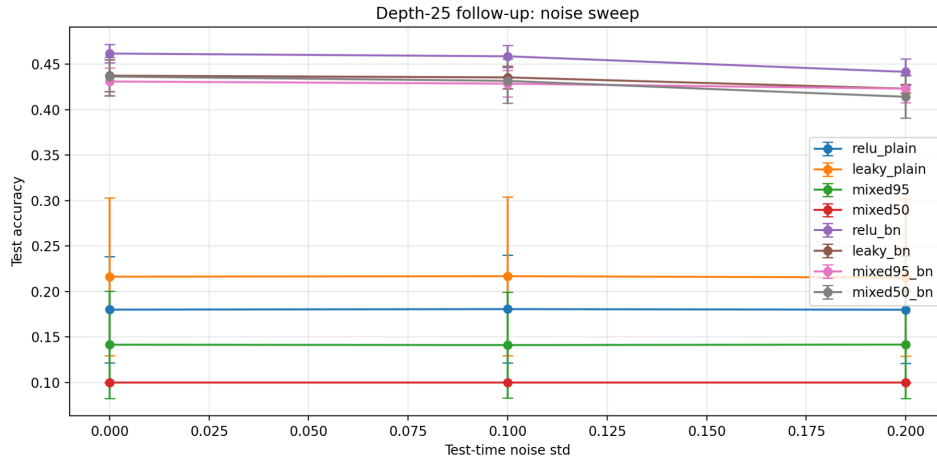


Figure 16: Depth-25 follow-up: test accuracy under added input noise. The relative ranking changes little across noise levels 0.0–0.2, and the normalized variants remain clearly stronger than the plain ones throughout.

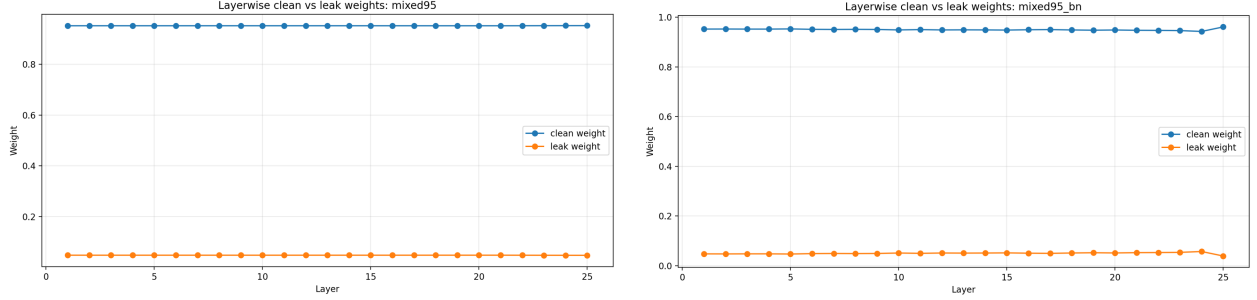


Figure 17: Layerwise clean versus leak weights for the mixed 95/5 initialization without batch normalization (left) and with batch normalization (right). In both cases the learned allocation stays strongly clean-dominant, so the main difference between failure and success is not a collapse of the path mixture itself.

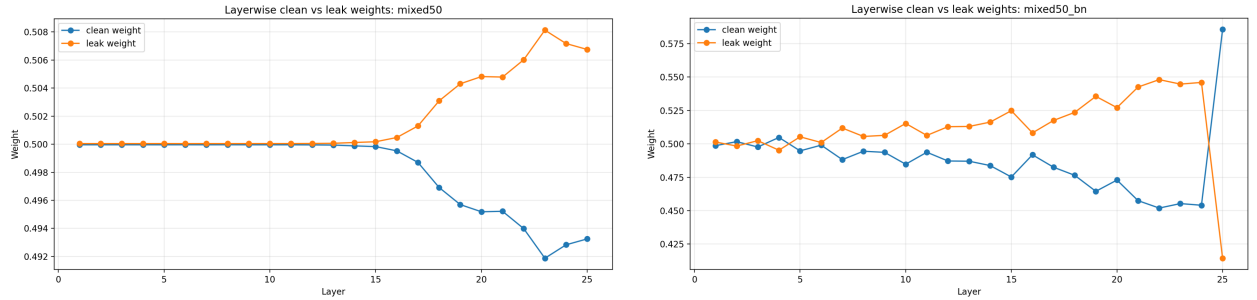


Figure 18: Layerwise clean versus leak weights for the mixed 50/50 initialization without batch normalization (left) and with batch normalization (right). The unnormalized model remains near parity and fails, whereas the normalized model supports a broader learned redistribution while still training to competitive accuracy.

D.4 MNIST label-corruption stress test

As a final supplementary check, we ran a CPU-friendly label-corruption stress test on MNIST using a three-hidden-layer multilayer perceptron with widths 512, 256, and 128. We compared ReLU, Leaky ReLU, Mixed Dual-Path Rattle with a roughly 95/5 initialization, and Mixed Dual-Path Rattle with a 50/50 initialization over three seeds and five epochs. The training labels were corrupted at rates 0.0, 0.3, 0.5, and 0.7, while evaluation used the clean validation and test labels. This experiment should be read as a targeted corruption stress test rather than as a new headline benchmark.

The main pattern is straightforward. On clean MNIST, all four activations were essentially tied near 0.96 test accuracy. As label noise increased, the mixed 50/50 model degraded more slowly than the others. At noise 0.3, it reached mean test accuracy slightly above the other three variants. At noise 0.5, the gap widened: mixed 50/50 was ahead of Leaky ReLU, ReLU, and mixed 95/5. At noise 0.7, mixed 50/50 remained best, with the other methods trailing behind.

Two additional diagnostics are noteworthy. First, the dead-rate pattern remained benign for all non-ReLU variants: Leaky ReLU and both mixed models stayed at zero under the chosen threshold

across all tested noise levels, while ReLU retained a small but nonzero dead rate. Second, the mixed 50/50 model trained with substantially smaller gradient norms than the other activations throughout the sweep. That behavior is consistent with a more damped optimization profile under severe label corruption, although this experiment alone does not establish the mechanism.

The mixed-path diagnostics make the comparison more specific. The mixed 95/5 model stayed strongly clean-dominant across all layers and all noise levels, with clean weight near 0.95 and leak weight near 0.05, and it did not show a corruption advantage over the standard baselines. By contrast, the mixed 50/50 model maintained a substantially larger auxiliary contribution, with clean and leak weights staying close to parity across layers. In this particular MNIST corruption setting, that more balanced allocation was associated with the strongest performance at the higher noise levels.

This supplementary result is useful because it identifies a narrower regime in which the framework may matter more: severely corrupted supervision in a simple dense network. At the same time, it should still be interpreted conservatively. It does not show a universal robustness advantage, and it does not imply that larger leak is generally better. What it does show is that under this specific label-corruption stress test, the mixed formulation with a much less clean-dominant initialization remained competitive on clean data and became the strongest of the tested activations as the label noise grew extreme.

D.5 MNIST extreme label-corruption extension

We also extended the same MNIST setup to even harsher label-corruption rates of 0.8 and 0.9, using ten epochs on CPU. This extension is useful because it helps distinguish two regimes: one in which the task is still difficult but learnable, and another in which the supervision becomes so corrupted that all methods are pushed close to chance behavior.

At noise 0.8, the mixed family remained competitive. Mean test accuracy was high for all methods, with the mixed variants remaining in the same general range as ReLU and Leaky ReLU. At noise 0.9, the problem is close to broken for every activation. Mean test accuracy falls near chance for all four methods. In other words, this regime does not provide an honest accuracy-win story for the mixed formulation. Its main value is diagnostic.

That diagnostic signal is still informative. ReLU accumulates large dead-unit rates as the corruption becomes extreme, while Leaky ReLU and both mixed variants stay at zero under the chosen threshold. The mixed models also retain smaller mean gradient norms than the standard baselines, especially for the 50/50 initialization. Those patterns are consistent with the leak-aware family acting as a more stability-oriented activation under severe corruption, even when that stability does not always translate into higher accuracy.

The layerwise mixtures sharpen that interpretation. Mixed 95/5 stays strongly clean-dominant at both extreme noise levels, with weights remaining near 0.95/0.05 across all three layers. Mixed 50/50 behaves differently: at noise 0.8 it stays moderately clean-dominant, whereas at noise 0.9 it shifts to a leak-heavier allocation. That shift is interesting as a learned response to a near-broken training signal, but the poor accuracy at 0.9 means it should be treated as a diagnostic observation rather than as evidence of successful adaptation.

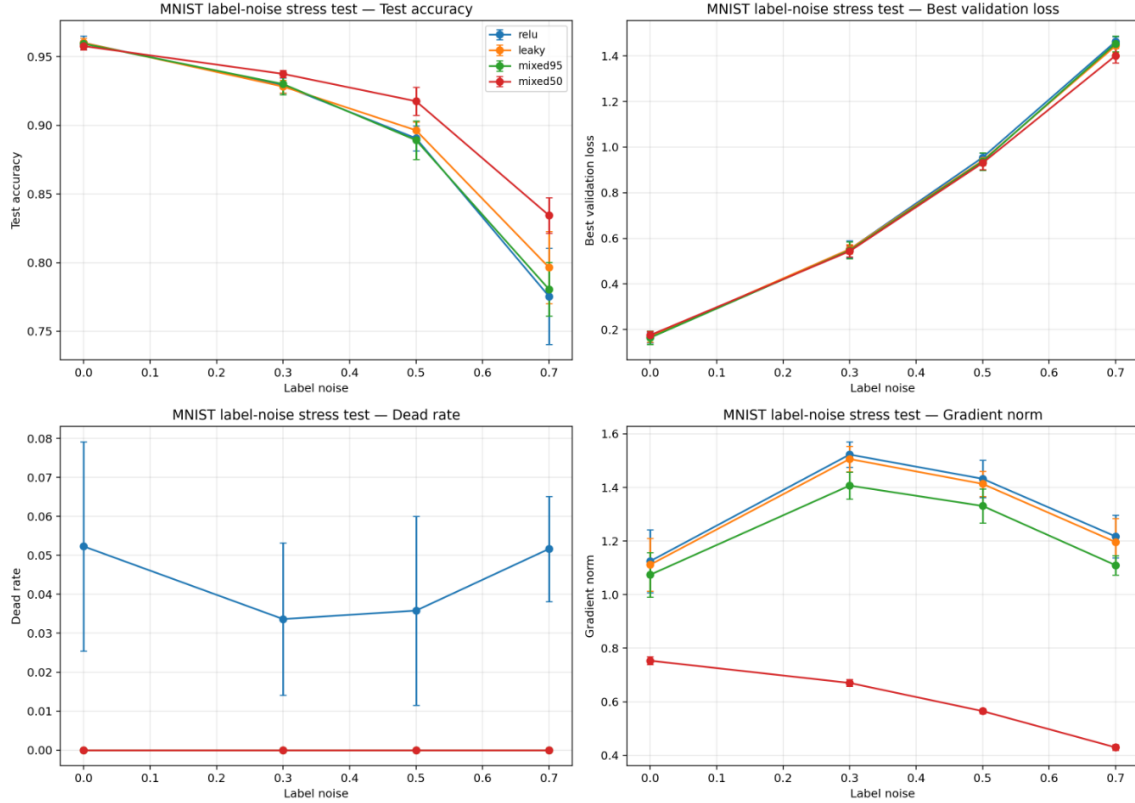


Figure 19: Supplementary MNIST label-corruption stress test over three seeds. The top-left panel shows test accuracy as label noise increases from 0.0 to 0.7. The top-right panel shows best validation loss. The bottom-left panel shows dead rate, which remains at zero for Leaky ReLU and both mixed variants under the chosen threshold. The bottom-right panel shows mean gradient norm, where the mixed 50/50 model follows a noticeably smaller-magnitude trajectory across the sweep.

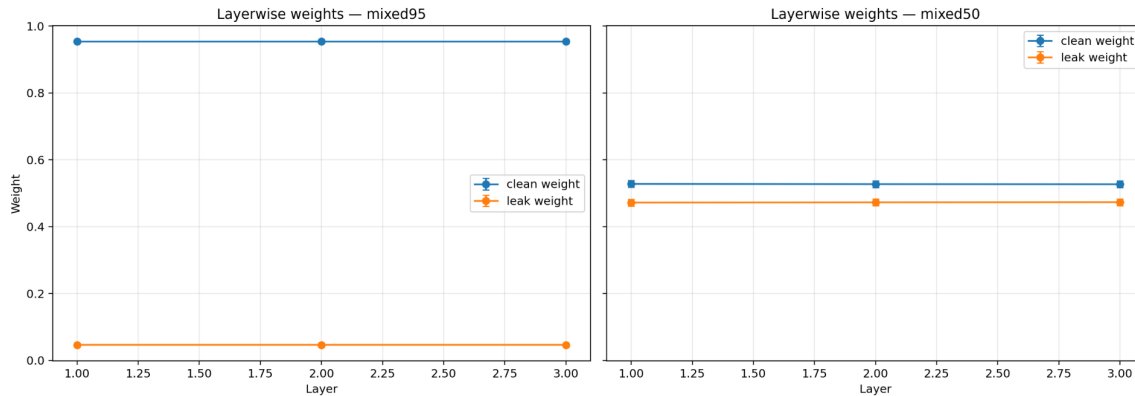


Figure 20: Layerwise clean versus leak weights for the two mixed variants in the MNIST label-corruption stress test. The mixed 95/5 model remains strongly clean-dominant across all three hidden layers, whereas the mixed 50/50 model keeps a substantially larger auxiliary contribution. In this experiment, the more balanced mixture is the one associated with the strongest performance at high label-noise rates.

Taken together with the earlier MNIST sweep, this extension supports a narrower and more defensible claim. Under severe label corruption, the leak-aware family can remain trainable and can sometimes match or exceed the simpler baselines, but the advantage is regime-dependent. The clearest pattern that survives both sweeps is not universal superiority; it is that the mixed family preserves controlled behavior and avoids the dead-unit collapse that becomes visible for ReLU in the most corrupted setting.

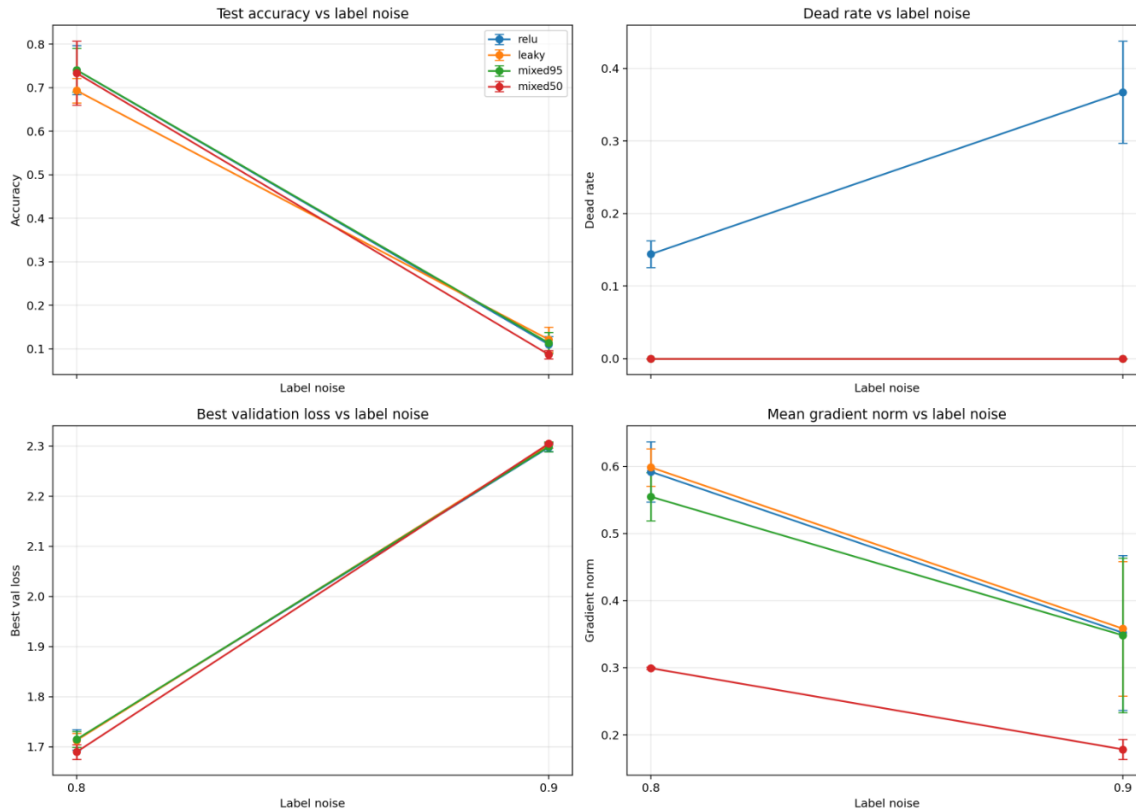


Figure 21: Extension of the supplementary MNIST corruption test to label-noise rates 0.8 and 0.9. At noise 0.8, the mixed variants remain competitive with the baselines, but at noise 0.9 all methods fall close to chance accuracy. The dead-rate panel highlights the clearest stability difference: ReLU accumulates substantial inactive units, whereas Leaky ReLU and both mixed variants remain at zero under the chosen threshold.

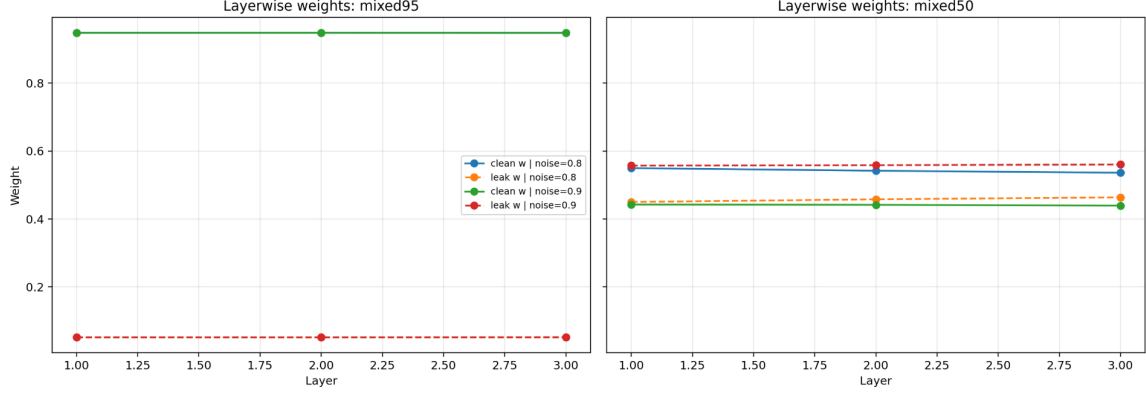


Figure 22: Layerwise clean versus leak weights for the two mixed variants at label-noise rates 0.8 and 0.9. Mixed 95/5 remains strongly clean-dominant, while mixed 50/50 shifts from a mildly clean-dominant mixture at noise 0.8 to a leak-heavier allocation at noise 0.9. Because the 0.9 regime is near chance for all methods, that latter pattern is best read as a diagnostic response rather than as a successful robustness mechanism.